

LYCANS AERODESIGN TEAM

Trainee Assignment

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Avionics System Design

This documentation is on choosing the right components for a standard quadcopter UAV (Unmanned Aerial Vehicle) drone used for photography.

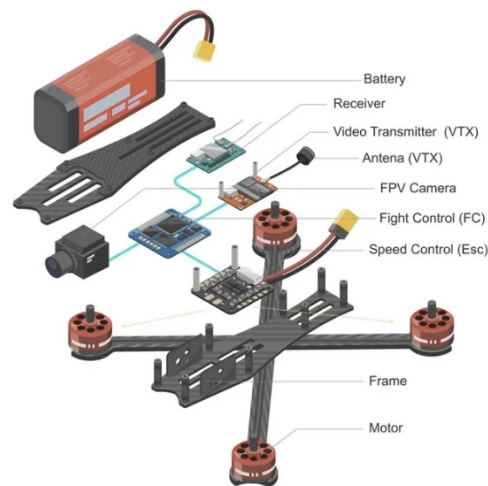


Figure 1: [drone parts](#)

- **Flight controller (FC):** drone brain, has built-in gyroscope and accelerometer for stability, speed control.
- **Receiver:** The receiver picks up control signals from your radio transmitter.
- **ESC:** converts the FC's low-power signals into the high-current power that brushless motors need.
- **GPS:** Important for automated navigation, like waypoint tracking and return-to-home.
- **FPV camera:** feeds video to the VTX.
- **VTX:** sends the live camera feed to your ground control station.
- **Magnetometers:** assist with navigation by detecting the Earth's magnetic field, improving directional accuracy.
- **Barometers:** estimate altitude by detecting air pressure, enabling precise height control and smoother landings.

0.1 Schematic

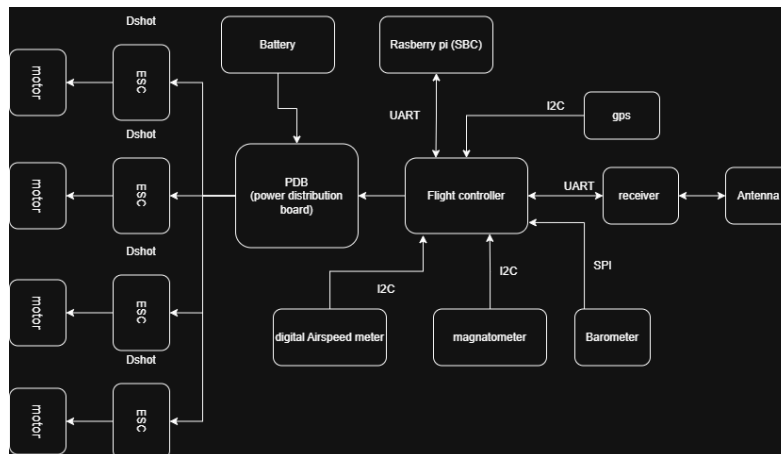


Figure 2: Aircraft design block diagram

SmartAP Power Distribution Board (PDB) is used to allow transferring the power from the battery to ESCs / Motors and generate power supply for the autopilot and other peripherals with different voltage levels.

0.2 Sensors and Protocols

Here are all sensors usable with the flight controller connections and protocol mentioned:

Connections to Flight controller

- **Receiver RX (UART):** Connects to the FC serial port, ExpressLRS (ELRS) receiver, which uses UART protocol and is bi-directional [Receivers](#).
- **GPS Module (UART):** Transmits position and navigation data to the FC via high-speed serial UART.
- **Magnetometer / Compass (I2C):** Measures heading orientation using a 2-wire I2C bus suitable for low-data-rate sensors.
- **Barometer (SPI / I2C):** Measures atmospheric pressure for altitude estimation; interfaced internally via fast SPI or I2C.
- **Telemetry Radio (UART):** Provides a two-way serial link via UART for real-time flight data and GCS communication.
- **SBC Companion Computer (UART):** Communicates high-level mission commands and video data with the FC using the MAVLink protocol.
- **ESC / Motors (PWM / DShot):** Transmits motor speed commands directly from the FC output channels.

0.3 Component Selection

Table 1: Component Selection Details

Component	Type	Reason of Choice	Amp Rating	Price	Weight
Battery	Li-ion 4S2P (Molicel P42A)	Higher energy density for longer endurance	4200 mAh (45A cont.)	1,600 EGP	580 g
Flight Controller	Pixhawk 4	Native ArduPilot support	800 mA	2,400 EGP	49 g
SBC	Raspberry Pi	High-level control	1200 mA	3,100 EGP	10 g
GPS	Holybro M9N	Built-in compass	200 mA	2,700 EGP	36.8 g
Receiver	ExpressLRS	Low latency and long range	1000 mA	991 EGP	0.4 g
Telemetry	SiK Telemetry	Real-time GCS link	100 mA	1,800 EGP	11.5 g
Fuse	Maxi Blade Fuse (Fast-Blow)	Protects avionics and battery from short circuits	40 A rating	50 EGP	3 g
Fuse holder	Inline Water-proof	Heavy-duty AWG12 leads for secure mounting	Up to 60 A	75 EGP	12 g

NOTE: The firmware that runs on the Flight controller (FC) is according to the FC type Ex: **Betaflight**, for any microcontroller, while **INAV** and **ArduPilot**, for certain microcontroller families.



T-Motor Velox V2306 V3.0 1
 2306 2550Kv Motor 2
 \$18.99 Out of stock \$

Figure 3: Velox-V2207 motor

- [how to choose suitable FC ?](#)
- [How to Choose ExpressLRS Receiver?](#)

1 Battery Calculations and Flight Time

1.1 Current Consumption Breakdown

By referring to Table 1, the overall current drawn from the system components is:

- FC = 800 mA = 0.8 A
- Raspberry Pi = 1200 mA = 1.2 A
- GPS = 200 mA = 0.2 A
- Receiver = 1000 mA = 1.0 A
- Telemetry = 100 mA = 0.1 A
- Avionics (FC + Raspberry Pi + GPS + Receiver + Telemetry) = $0.8 + 1.2 + 0.2 + 1.0 + 0.1 = 3.3$ A
- 4 × Velox Motors (Cruise/Hover draw) = 4×3.0 A = 12.0 A
- **Total Cruise Current Draw (I_{total}):** 3.3 A + 12.0 A = 15.3 A

1.2 Battery Sizing and Flight Time Estimation

Selected Battery: **Li-ion 4S2P (4200 mAh, 14.8 V, 45 A Continuous Output)**

- **Flight Time (T_{flight}):**

$$T_{\text{flight}} = \frac{\text{Battery Capacity (Ah)}}{I_{\text{total}}} \times 60 = \frac{4.2 \text{ Ah}}{15.3 \text{ A}} \times 60 \approx 16.4 \text{ minutes} \quad (1)$$

- **Discharge Safety Check:** The battery continuous discharge capability is 45 A, which easily covers the continuous cruise requirement (15.3 A).

1.3 Fuse Selection

A **40 A Blade Fuse** was chosen to protect the system against short circuits, while safely handling the normal 15.3 A cruise current. “[cite: 1]

2 Resources

- [Overview on drone components choice](#)
- [Overview on Quadcopter hardware components](#)
- [ArduPilot Pixhawk Power Wiring Overview](#)